

VICTOR MIGUEL VELAZQUEZ ESPITIA

Mexico City | rotciveb@gmail.com | +39 3475466543 | [linkedin.com/in/victor-velazquez-intelligent-engineering/](https://www.linkedin.com/in/victor-velazquez-intelligent-engineering/) | <https://github.com/victor-velazquez-ai>

PROFILE

Ambitious and technically skilled AI engineer with experience in Agentic AI, Robotics, Quant, and back-end systems. Eager to contribute to high-tech teams by leveraging deep mathematical understanding, wide technical knowledge, strong coding abilities, and analytical thinking.

PROFESSIONAL EXPERIENCE

AI Engineer, El Paso Labs, Remote

Jun 2025 - May 2026

- Architected and developed agentic AI systems to automate business workflows, connecting LLMs with operational and enterprise platforms.
- Built production AI applications, backend services, and APIs for AI-driven task execution, structured data processing, and multi-step workflow automation.
- Developed from scratch a repeatable observability infrastructure for agentic workflows using Langfuse, standardizing tracing, monitoring, and evaluation across production AI systems.
- Designed integrations with Slack, Box, Front, DocuSign, Airtable, Notion, Gmail, NetSuite, and external APIs to enable end-to-end intelligent workflows.
- Developed a custom enterprise-level MCP service with authentication and authorization integration, enabling secure and reusable connections between AI agents and business systems.

AI Engineer (Part-time), A2C, Remote

Nov 2025 - May 2026

- Specialized in Amazon Bedrock and AWS-based AI projects, contributing to backend systems and production-oriented agentic workflow development.
- Worked as an AI assessment engineer to identify opportunities for agentic workflows, system optimization, and practical business adoption.
- Contributed to the Assistments project, developing a summarizing agent and educational performance analyzer for student insights and reporting.
- Implemented guardrails, prompt management workflows, model evaluation pipelines, and testing strategies for optimization, reliability, and model selection. Amazon CloudWatch for agentic and system observability, debugging, and performance monitoring in AWS environments.

Software Engineer, Dynamic Optics, Padova

Sep 2024 - Jul 2025

- Developer for a software library for Adaptive optics operation. Achieved the control of the Satellite to Telescope laser communication application under high atmospheric turbulence conditions.
- Implemented an embedded system thermal control scheme to supervise the conditions of a whole Satellite system, guaranteeing safe station operation.
- Team Manager and Developer for an end-to-end control software library and GUI solution for a 6 DOF robotic arm for task execution, peripheral devices control, and actuation.

Research Traineeship, University of Padova

Jan 2024 - Sep 2024

- Developed a medical assistance Deep Learning Segmentation model for polyps identification, including PyTorch model development, training, testing, and tuning.
- Implemented innovative data augmentation and transfer learning strategies due to limited data availability, increasing accuracy by 5%.
- Implemented an ensembling unified architecture to enhance generalization capabilities, surpassing SOTA accuracy in 3 out of 5 test datasets.

Robotics Software Engineer, Subdron, Bregenz

Jan 2023 - Jan 2024

- Implemented an underwater mapping algorithm based on 3D point clouds for damage and condition inspection, with accuracy above 90% for relevant object detection.

- Performed ROS data engineering for error identification during robot operations by automatically generating and querying pandas data frames.
- Optimized point cloud processing, smoothing, and navigation accuracy by addressing unpredictable noise, improving performance by 20%.

Quant & Portfolio Analyst, LatamLead

2022 - 2023

- Designed and implemented systematic strategies based on Modern Portfolio Theory, optimizing asset selection with Omega Ratio and Sharpe Ratio filters.
- Developed trend-following systems using Python and back-tested them with custom-built frameworks and TradingView Pine Script.
- Translated macroeconomic insights into quantitative inputs for strategy calibration, including CPI, PMI, and rate-hike scenarios.

Machine Learning Engineer, Natma, Mexico City

Jan 2020 - Sep 2022

- Designed and deployed a forecasting model using neural networks and regression to optimize seasonal sales strategies, resulting in a 10% revenue increase.
- Built computer vision pipelines using YOLO, OpenCV, and TensorFlow, deployed on edge devices for real-time performance.
- Built and deployed a chatbot system using AWS Lambda, Twilio API, and WhatsApp Business API, managing asynchronous event processing via RESTful endpoints.
- Led the design and implementation of an autonomous robotic platform using SLAM for real-time localization and mapping with Python and ROS.

EDUCATION

University of Padova

Feb 2022 - Jul 2024

Master's degree in Computer Engineering

Acamica Careers

Jan 2021 - Dec 2021

Data Science Nano Degree

Tecnological Institute of Monterrey (CDMX)

Aug 2016 - Dec 2020

Bachelor's degree in Mechanical Electrical Engineering

Universidad Politecnica de Catalunya

Aug 2018 - Aug 2019

International Exchange in Electrical Engineering

TECHNICAL SKILLS

Python | C++ | C | PyTorch | Pandas | AWS | Bedrock | Scikit-learn | Deep Learning | Spark | R | Docker | Git | Jira | OpenCV | Gazebo | ROS | Linux | MATLAB | Simulink | SQL | Tableau | Keras | TensorFlow | FastAPI | Flask | CrewAI | GPT | Chroma | Pinecone | LangGraph | LangChain | Azure | React

COURSES

LangChain AI Developer MasterClass, Udemy 2025 | SQL for Data Science, University of California 2023 | Self Driving Cars Specialization, University of Toronto 2023 | Algorithmic Trading Python + AWS, Udemy 2022 | AWS Fundamentals, Amazon 2021 | Natural Language Processing, Deep Learning AI 2020 | Deep Learning Specialization, Deep Learning AI 2020 | Deep Reinforcement Learning, Super Data Science 2021

ADDITIONAL INFORMATION

Languages: Spanish Native, English Fluent, Italian Proficient, German Beginner